Word Embedding Analysis Report

SVD · Word2Vec (CBOW) · GloVe

POS Tagging · Analogy Tests · Bias Analysis

1. SVD Hyperparameters

The SVD embedding uses a dimension of 350, applied to vocabulary words with a minimum frequency of 10.

2. Analogy Test

Three embedding variants are evaluated using the analogy formula:

```
vec(B) - vec(A) + vec(C)
```

The top-5 nearest neighbours (excluding query words) are selected by cosine similarity.

2.1 SVD

Paris : France :: Delhi : ? (capital city)

⚠ *One or more query words not in vocabulary — result unavailable.*

King : Man :: Queen : ? (gender)

Rank	Word	Cosine Similarity
1	woman	0.7993
2	like	0.7855
3	really	0.7818
4	gone	0.7818
5	little	0.7812

Swim : Swimming :: Run : ? (verb tense)

Rank	Word	Cosine Similarity
1	put	0.7834
2	time	0.7800

3	said	0.7779
4	gone	0.7756
5	coming	0.7734

2.2 Word2Vec (CBOW)

Paris : France :: Delhi : ? (capital city)

Rank	Word	Cosine Similarity
1	properties	0.9781
2	unity	0.9771
3	sides	0.9768
4	connected	0.9767
5	inner	0.9764

King : Man :: Queen : ? (gender)

Rank	Word	Cosine Similarity
1	whether	0.9978
2	every	0.9978
3	considered	0.9978
4	during	0.9978
5	these	0.9978

Swim : Swimming :: Run : ? (verb tense)

Rank	Word	Cosine Similarity
1	after	0.9990
2	good	0.9990
3	days	0.9989
4	they	0.9989
5	has	0.9989

2.3 GloVe

Paris : France :: Delhi : ? (capital city)

Rank	Word	Cosine Similarity
1	india	0.8602
2	pakistan	0.7835
3	lanka	0.6693
4	bangladesh	0.6641
5	sri	0.6440

King : Man :: Queen : ? (gender)

Rank	Word	Cosine Similarity
1	woman	0.8040
2	girl	0.7349
3	she	0.6818
4	her	0.6592
5	mother	0.6542

Swim : Swimming :: Run : ? (verb tense)

Rank	Word	Cosine Similarity
1	three	0.7016
2	running	0.7008
3	since	0.6961
4	four	0.6906
5	leading	0.6884

3. Gender Bias in Pre-trained Embeddings

Cosine similarity between occupation words and gendered words (man / woman) computed on GloVe pre-trained embeddings.

Occupation	cos(word, man)	cos(word, woman)	Closer To
doctor	0.6092	0.6333	**woman (small diff.)**
nurse	0.4562	0.6139	**woman**
homemaker	0.2356	0.4258	**woman**

4. Frozen Embeddings

For all three cases, the embedding layer is frozen to leverage the classifier for prediction. This provides a fair implicit comparison across different embedding models using the same classifier architecture.

5. Embedding Comparison

Model	Accuracy	Macro-F1
SVD	0.90	0.83
CBOW	0.73	0.59
GloVe	0.96	0.91

Among all embedding models, GloVe performs best. This is attributed to its training on a larger dataset, providing broader context and richer co-occurrence statistics.

5.1 Confusion Matrix — Best Model (GloVe)

6. Hyperparameter Analysis

Model	Window Size	Hidden Layer	Learning Rate	Accuracy
SVD	4	256	1e-3	0.87
SVD	4	512	1e-3	0.87
CBOW	4	256	1e-3	0.71
CBOW	4	512	1e-3	0.71
GloVe	4	256	1e-3	0.96
GloVe	4	512	1e-3	0.97

7. False Predictions

Sample misclassifications from the GloVe POS tagger — first 5 errors on the test set.

#	Context Window	True Tag	Predicted Tag
1	eileen got to dancing ,	PRT	ADP
2	little tiny dancing step to	VERB	NOUN
3	to a hummed tune that	VERB	ADV
4	could hardly notice , and	VERB	NOUN
5	to pick up strange men	ADP	PRT